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HYBRID MEMORY REPLAY: BLENDING REAL AND DISTILLED DATA FOR CLASS INCREMENTAL LEARNING

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ABSTRACT

Incremental learning (IL) aims to acquire new knowledge from current tasks while retaining knowledge learned from previous tasks. Replay-based IL methods store a set of exemplars from previous tasks in a buffer and replay them when learning new tasks. However, there is usually a size-limited buffer that cannot store adequate real exemplars to retain the knowledge of previous tasks. In contrast, data distillation (DD) can reduce the exemplar buffer's size, by condensing a large real dataset into a much smaller set of more information-compact synthetic exemplars. Nevertheless, DD's performance gain on IL quickly vanishes as the number of synthetic exemplars grows. To overcome the weaknesses of real-data and synthetic-data buffers, we instead optimize a hybrid memory including both types of data. Specifically, we propose an innovative modification to DD that distills synthetic data from a sliding window of checkpoints in history (rather than checkpoints on multiple training trajectories). Conditioned on the synthetic data, we then optimize the selection of real exemplars to provide complementary improvement to the DD objective. The optimized hybrid memory combines the strengths of synthetic and real exemplars, effectively mitigating catastrophic forgetting in Class IL (CIL) when the buffer size for exemplars is limited. Notably, our method can be seamlessly integrated into most existing replay-based CIL models. Extensive experiments across multiple benchmarks demonstrate that our method significantly outperforms existing replay-based baselines.

1 Introduction

Incremental Learning (IL) [35, 10, 7, 36] is an emerging technique that emulates the human ability to continuously acquire new knowledge in an ever-changing world. However, the key challenge is that IL always suffers from *catastrophic forgetting* [21], which means the model drastically forgets previously acquired information upon learning new tasks. Thus, how to mitigate catastrophic forgetting in IL remains a significant challenge.

In this work, we primarily focus on a particularly challenging scenario known as class incremental learning (CIL) [6, 45, 32, 31]. CIL aims to learn new and disjoint classes across successive task phases and then accurately predicts all classes observed thus far in each phase, without prior knowledge of task identities. A common approach to enhance CIL performance is the use of replay, which stores a subset of exemplars from old tasks and revisits them when learning new tasks. Storing real exemplars can significantly enhance the performance of CIL [25, 5]; however, the performance will be affected by the limitation of the exemplar buffer size. To decrease the dataset size, recent studies have explored data distillation (DD) [19, 8] to distill compact *synthetic exemplars* from real datasets. However, [38] indicates that the effectiveness of synthetic exemplars declines as their quantity increases. Motivated by these findings, we aim to develop a new hybrid memory that combines the advantages of both real and synthetic exemplars for replay-based CIL methods.

In this paper, we introduce a novel hybrid memory that optimizes a limited number of both synthetic and real exemplars for replay-based CIL methods. A key challenge is enabling these limited exemplars to capture adequate information from data samples of previous tasks. To address this challenge, we aim to introduce data distillation (DD) into CIL, which condenses large datasets into a small set of representative samples. However, classical DD techniques are not directly applicable to replay-based CIL due to the requirement of checkpoints on multiple training trajectories. To address this challenge, we first propose an innovative Continual Data Distillation (CDD) method that adapts DD for replay-based CIL, extracting informative synthetic exemplars from previous tasks' datasets. Interestingly, experiments in Sec. 3 across different limited exemplar buffer sizes reveal that while memory relying solely on synthetic exemplars initially outperforms memory using only real exemplars, its performance becomes less effective than the memory only with real exemplars as the buffer size increases. To address this issue, we further develop conditional real data selection that chooses optimal real samples to complement synthetic data. As a result, the interplay between selected real exemplars and synthetic data enhances model performance within a limited exemplar buffer size. As illustrated in Fig. 1, our proposed hybrid memory significantly outperforms the original real exemplars across various replay-based CIL methods. Finally, in Sec. 5, we evaluate our approach by integrating it into several existing replay-based CIL models and compare their performance against baseline models. The results validate that our hybrid memory consistently leads to superior performance.

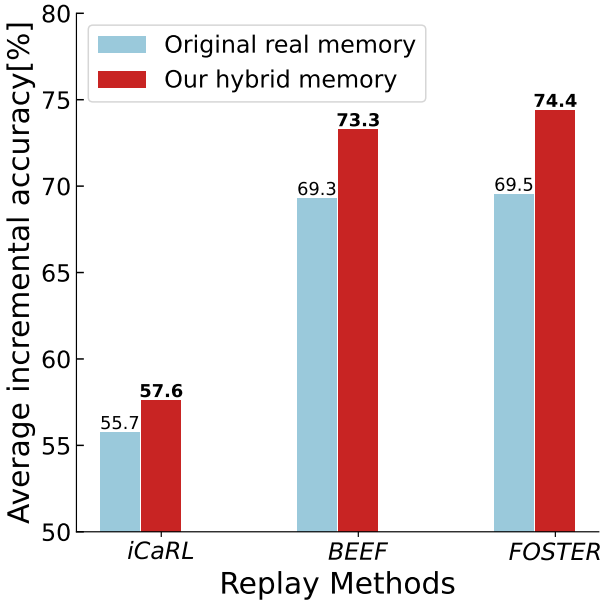

Figure 1: Performance comparison between the real memory as used by the original CIL methods and our hybrid memory across multiple baselines: iCaRL [25], BEEF [31], and FOSTER [32], all with the same exemplar buffer size on CIFAR-100 [14].

Our contributions are four-fold: 1) to our best knowledge, we are the first to synergize limited synthetic and real exemplars to boost replay-based CIL performance; 2) we develop a novel conditional real data selection that optimally chooses real exemplars, which can complement the synthetic data effectively; 3) our approach can be inserted into many existing replay-based CIL models to improve their performance; and 4) extensive experiments demonstrate that our method can significantly outperform existing replay-based CIL approaches with a limited number of saved exemplars.

2 Related Work

Class Incremental Learning. Existing works of CIL can be categorized into three types: regularization-based methods, architecture-based methods, and replay-based methods.

(i) ***Regularization-based methods*** seek to mitigate catastrophic forgetting by incorporating explicit regularization terms to balance the knowledge from old and new tasks [17, 13, 15, 18, 1]. However, only using regularization-based methods often proves inadequate for preventing catastrophic forgetting. Therefore, these methods often collaborate with the replay-based methods to boost performance. (ii) ***Architecture-based methods*** try to mitigate catastrophic forgetting by either dynamically expanding the model's parameters to accommodate new tasks [20, 1, 7] or by learning new tasks in parallel using task-specific sub-networks or modules [28, 9, 12]. However, these approaches often require substantial memory to store the expanded network or multiple sub-networks and typically depend on task identities to determine the appropriate parameters or sub-networks for a given task. Consequently, their applicability is significantly limited. (iii) ***Replay-based methods*** are designed to mitigate catastrophic forgetting by preserving a small subset of old task exemplars for replay while learning new tasks. One line of these methods [25, 6, 32, 31] maintains a limited memory for storing real exemplars from old tasks. However, due to data privacy concerns and inherent memory constraints, the number of real exemplars stored is typically small, significantly affecting performance in applications. Another line [30, 22, 16, 26] employs generative models, such as Variational Autoencoders (VAE) and Generative Adversarial Networks (GAN), to create synthetic exemplars of old tasks for replay. However, these methods often face challenges with label inconsistency during continual learning [2, 22], and is closely related to continual learning of generative models themselves [33]. In contrast, this paper aims to condense dataset information into a few synthetic exemplars without training additional models.

Data Distillation. Data distillation (DD) aims to condense the information from a large-scale dataset into a significantly smaller subset. This process ensures that the performance of a model trained on the distilled subset is comparable to that of a model trained on the original dataset. Existing DD methods can be grouped into three categories based on the optimization objectives: *performance matching* [19, 44], *parameter matching* [3, 39, 8], and *distribution matching* [40, 29]. While a few studies [40, 29] have applied DD to a simple specific IL application like GDumb [24], they cannot be directly applied to the general replay-based CIL methods. This is because traditional DD techniques require training *a set of models* randomly sampled from an initialization distribution. However, for typical replay-based CIL, the initial model parameters for each task are derived from the weights trained on the previous task. Thus, directly applying existing DD techniques to CIL introduces significant computational costs, as it requires training an additional set of models to get multiple trajectories. To address this issue, we introduce the Continual DD (CDD) technique to suit replay-based CIL.

3 Why Hybrid Memory?

In this section, we mainly introduce the rationale behind the need for hybrid memory in CIL by comparing real, synthetic, and bybrid memory.

To address catastrophic forgetting in CIL, general replay-based methods attempt to store real exemplars from previous tasks in a limited exemplar buffer. Intuitively, saving high-quality exemplars from old tasks can enhance the performance of replay-based CIL. However, a size-limited exemplar buffer cannot store adequate real exemplars to retain the knowledge of previous tasks. To address this challenge, we introduce a Continual Data Distillation (CDD) to compress important information from each task's dataset into a limited number of synthetic exemplars, replacing the need to store real exemplars in the buffer.

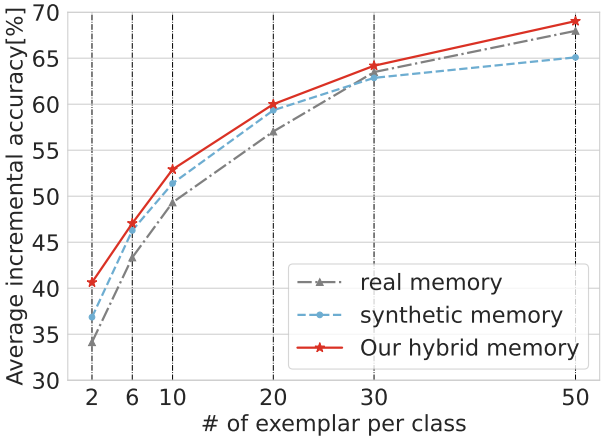

Figure 2: Performance evaluation of iCaRL [25] using different exemplar buffer sizes for real memory, synthetic memory, and our hybrid memory. "real memory" refers to buffers containing only real exemplars selected by iCaRL. "synthetic memory" contains only synthetic exemplars generated by CDD.

However, synthetic exemplars alone are not always sufficient. As their number increases, performance tends to plateau due to the inevitable information loss that occurs during the distillation process. As shown in Fig. 2 and Appendix E, we evaluate various replay-based CIL methods with buffers containing either only real exemplars or only synthetic exemplars generated by CDD across different buffer sizes on CIFAR-100 [14]. We can observe that synthetic memory performs better than real memory when the buffer size is small, suggesting that synthetic exemplars capture more information per exemplar when storage is constrained. However, as the buffer size increases, synthetic exemplars become less effective than real exemplars, likely due to the inherent important information loss in the distillation process. Motivated by these observations, we propose to construct a hybrid memory that combines the conditionally selected real exemplars with the synthetic ones to complement each other. In the following Sec. 4, we will detail the proposed method.

4 Proposed Hybrid Memory

In this section, we first present the problem of using hybrid memory in replay-based CIL. Then we detail the proposed hybrid memory to mitigate the catastrophic forgetting in CIL.

4.1 Problem Formulation

This work aims to build a high-quality hybrid memory to mitigate the catastrophic forgetting problem in CIL, which involves sequentially learning distinct tasks, each consisting of disjoint groups of classes. Specifically, the real dataset for task-t is denoted as $\mathcal{R}_t \triangleq \{(x_i, y_i)\}_{i=1}^{n_t}$, where y_i belongs to the label space C_t of task-t, and n_t represents the number of instances. Unlike traditional replay-based methods that store only limited real exemplars, our approach maintains a hybrid memory with a limited number of hybrid exemplars from learned tasks' classes. The hybrid memory is represented as $\mathcal{H}_{1:t-1} \triangleq \bigcup_{i=1}^{t-1} \mathcal{H}_i$, where $\mathcal{H}_t \triangleq \{\mathcal{S}_t, \mathcal{R}_t[A_t]\}$ represents the hybrid memory for task-t. Here, $\mathcal{S}_t$